

GNSS-RTK Cadastral Measuring

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Abstract: GNSS-RTK cadastral measuring refers to the application of Global Navigation Satellite System (GNSS) technology—specifically Real-Time Kinematic (RTK) methods—in cadastral measuring and surveying, which focuses on defining property boundaries and land ownership. This modern approach enhances traditional practice by providing real-time kinematic centimetre-level positioning for in-motion surveys, whereas static methods can achieve sub-centimetre (often millimetre-level) accuracy. The integration of GNSS-RTK has transformed cadastral measuring, shifting from manual and conventional techniques to digital solutions, and has become indispensable in land administration and management. While cadastral surveying encompasses the legal and administrative aspects of land ownership, cadastral measuring provides the technical foundation that ensures precise spatial data collection, which is critical for legal boundary determinations, land registration and infrastructure planning.

The significance of GNSS-RTK in cadastral measuring is reflected in its widespread adoption across land surveying, agriculture and urban planning, where demand for high-precision geospatial data continues to grow in increasingly complex spatial environments. The technology enables accurate boundary determination for legal transactions, infrastructure development, and environmental monitoring. By streamlining workflows and reducing disputes over ownership, GNSS-RTK has become a cornerstone of modern cadastral applications.

Implementation is not without challenges. Signal obstruction in urban areas, multipath from reflective surfaces, and the initial cost of equipment can limit uptake, particularly for smaller firms. In addition, reliance on communications infrastructure raises concerns about accessibility in remote regions with limited network coverage. Addressing these issues is essential for the continued evolution and integration of GNSS-RTK in cadastral practice.

As the field progresses, ongoing advances in GNSS signal technology and the integration of RTK with emerging platforms, such as drones and autonomous systems, promise further gains in accuracy and efficiency. These developments highlight both the capacity for innovation in cadastral measuring and the role of GNSS-RTK as a foundational technology for the future of land administration and management.

To illustrate its practical application, this paper presents a case study demonstrating the implementation of GNSS-RTK technology in a real cadastral surveying project. The case study outlines the methodology, equipment used, and results obtained, offering insight into the effectiveness of GNSS-RTK for property boundary delineation. It also discusses both the benefits and limitations of this approach within the context of modern cadastral practices.

Keywords: GNSS, RTK, cadastral measuring, surveying, land administration, accuracy, positioning

Introduction

The evolution of cadastral measuring has been strongly influenced by advances in Global Navigation Satellite System (GNSS) technology, particularly the adoption of Real-Time Kinematic (RTK) techniques. Historically, cadastral marks served as primary reference points for boundary identification, area computation, and mapping updates; in countries such as Italy, these marks supported cartographic networks that underpinned land administration and registration systems (Timár, Baiocchi & Lelo, 2013). Earlier cadastral practice relied on conventional instruments, extensive field notes and physical landmarks, which constrained both productivity and the repeatability of measurements across large areas and extended timeframes.

The introduction of satellite positioning in the late twentieth century marked a significant change in cadastral work. GNSS was first deployed for navigation and then progressively adapted for precise positioning as receiver capabilities, signal availability, and processing methods improved. For cadastral tasks, this transition enabled surveyors to determine coordinates with higher accuracy and to establish reliable control in shorter field campaigns (Betzner, 2024). RTK further accelerated this transformation by providing centimetre-level accuracy in real time motion, through the use of a fixed reference, a mobile rover, and carrier-phase observations with differential corrections. This advancement has made it feasible to conduct cadastral measurements with a level of precision previously unattainable using traditional methods (Rodríguez-Alvarez, Muñoz-Martin & Morris, 2023).

Digitalisation has accompanied these technical gains. The spread of Continuously Operating Reference Stations (CORS) and the network delivery of corrections have increased reliability and supported common workflows for quality control, data exchange, and archiving. Low-latency links and standard protocols allow the distribution of correction data via mobile networks, facilitating the integration of GNSS technology with field software, geographic information systems (GIS), and asset databases. These capabilities have been incorporated into commercial and open-source tools, improving accessibility for both public-sector organisations and private practice (Gibblings & Zahl, 2014; Capua, 2018).

Despite these advances, limitations remain. Signal obstruction and multipath, in dense urban environments can interrupt phase tracking; connectivity may be intermittent in remote areas; and high capital and maintenance costs can limit adoption among smaller organisations. These constraints motivate continued development in receiver design, correction services, and multi-sensor integration, to maintain accuracy and availability when conditions are less than ideal. To demonstrate practical value, this study presents a case of GNSS-RTK applied to a cadastral measuring task at MCAST. The study describes the site, equipment, and procedures; reports on accuracy and repeatability results; and discusses how the observed performance aligns with contemporary cadastral requirements and operational constraints.

Literature Review

Cadastral measuring underpins land administration by supplying precise, auditable and spatial evidence for the definition of property boundaries and associated rights. Beyond parcel delineation, reliable measurement supports valuation, taxation, planning and tenure security, and contributes to broader objectives in governance and sustainable development (Williamson et al., 2010). Effective systems depend on clear legal frameworks, appropriate institutional arrangements and long-term maintenance of cadastral datasets, alongside the technical capacity to acquire, store and share high-quality geospatial information.

Despite its importance, cadastral measuring faces numerous challenges, including technical difficulties related to terrain, technological adaptation and regulatory hurdles such as boundary disputes and compliance with zoning laws (Williamson et al., 2010). The landscape is further shaped by institutional and economic factors, particularly in regions where traditional land management practices must evolve to contemporary needs. As such, the ability to navigate these challenges is crucial for surveyors and land administrators.

Recent advancements in technology, particularly the use of drones, GIS and artificial intelligence (AI), are transforming cadastral measuring, enabling more effective data analysis and enhanced decision-making capabilities. These innovations are anticipated to improve not only the accuracy and efficiency of land surveys but also the overall functionality of land administration systems (Bennett et al., 2020). The ongoing evolution of cadastral practices underscores their indispensable role in shaping land governance and sustainable development in the modern era.

Historical Background

Cadastral measuring and surveying have a rich history that dates back to ancient civilisations, reflecting humanity's enduring need to manage and document land ownership (Williamson et al., 2010). The practice of recording land boundaries can be traced to ancient Egypt, where rulers implemented surveying techniques as early as 3000 BC to allocate agricultural land and to monitor the Nile's flooding patterns, which necessitated the periodic re-establishment of boundaries (Clarke & Engelbach, 1990). Similarly, the Babylonians developed early cadastral systems using clay tablets to document property measurements and ownership, laying foundational principles for modern land administration (Dowson & Sheppard, 1952; Kain & Baignet, 1992). The Roman Empire significantly advanced cadastral practices by formalising land surveying as a profession. The Romans meticulously documented land distributions and established tax registers around 300 AD, which were essential for efficient governance and revenue collection (Dilke, 1971). Landmark examples such as the Domesday Book, compiled in 1086 under William the Conqueror, illustrate how historical cadastral surveys facilitated economic planning and land governance in medieval England (Kain & Baignet, 1992). The evolution of surveying techniques continued through the Renaissance, when innovations in measurement tools allowed for more precise mapping of land (Richeson, 1966). The nineteenth century marked a further turning point, particularly in Europe and North America, as national governments established comprehensive land registries that enhanced security of tenure and minimised boundary disputes through systematic records of ownership (Kain & Baignet, 1992). Today, modern cadastral practice incorporates sophisticated methods such as GPS/GNSS, GIS and LiDAR, alongside traditional techniques, to ensure accurate documentation and dynamic property management (Ghilani & Wolf, 2015). This continuing evolution underscores the critical role of cadastral surveying in facilitating land administration, economic development, and social stability.

Cadastral Measurement and Surveying

Cadastral surveying is a specialised branch of surveying dedicated to the establishment, re-establishment, and documentation of real property boundaries. It serves as a cornerstone in managing land resources, defining legal property boundaries and upholding land ownership frameworks, thereby guaranteeing precise delineation for ownership rights, tax assessments, and development planning. This process encompasses the physical delineation of boundaries as well as the determination of dimensions, areas, and associated rights for properties, irrespective of their location on land or water, or the presence of natural or artificial features (Enemark, 2005). The role of the cadastral surveyor is critical in this context, as practitioners must apply spatial measurement principles alongside relevant legal principles to respect neighbouring titles during the surveying process. Methodologies have evolved significantly, from manual triangulation

to advanced satellite-based positioning systems such as GNSS-RTK (Hofmann-Wellenhof, Lichtenegger & Wasle, 2008).

Cadastral surveying and cadastral measuring are closely related concepts, but they serve different roles within land administration. Cadastral measuring refers specifically to the technical process of obtaining precise spatial data related to property boundaries using GNSS-RTK, total stations, LiDAR and other geospatial technologies to determine distances, angles and coordinates. In contrast, cadastral surveying encompasses a broader scope by integrating cadastral measurement with legal and administrative processes to formally establish, document, and register land ownership. While measuring focuses on physical data collection, surveying ensures that these measurements comply with legal frameworks, property records, and national land administration policies. In essence, all cadastral surveys require cadastral measurements, but not all cadastral measurements lead to legal surveys, making surveying the final, legally binding step in land boundary determination (see Table 1).

Aspect	Cadastral Measuring	Cadastral Surveying
Definition	The technical process of obtaining precise spatial data for property boundaries.	A legal and administrative process integrating measurements with land ownership documentation.
Scope	Focuses on data collection, including distances, angles, and coordinates.	Includes measuring, legal validation, and official land registration.
Purpose	To provide accurate physical measurements for land surveys.	To establish legally recognised property boundaries and ownership records.
Methods Used	GNSS-RTK, total stations, LiDAR, photogrammetry, GIS.	Integrates measuring tools with legal property records and land registration systems.
Legal Aspect	Not legally binding; provides raw geospatial data.	Legally binding; used in property transactions and dispute resolution.
Output	Raw spatial data, GPS coordinates, and digital/physical boundary markers.	Cadastral maps, land registry documents, and official survey reports.
Users	Surveying engineers, geospatial analysts, and GIS specialists.	Licensed surveyors, land registry officials, government agencies.
Example Use Case	Measuring a land parcel using GNSS-RTK for mapping purposes.	Submitting an official land survey to a national land registry for ownership documentation.

Table 1: *Comparison of Cadastral Measuring and Cadastral Surveying*

Principles of GNSS-RTK Measurement

GNSS-RTK determines positions using a pair of receivers: a base station set on a control point of known coordinates and a rover that occupies the points of interest. The rover observes satellite signals and simultaneously receives real-time corrections from the base. By combining the two information streams, the rover obtains centimetre-level positions suitable for cadastral work (Hofmann-Wellenhof, Lichtenegger & Wasle, 2008).

The precision comes from using the carrier phase of the satellite signal which has a wavelength-scale sensitivity that is significantly finer than code-based measurements.

Carrier-phase observations, however, are susceptible to several sources of error. Typical contributors include satellite and receiver clock bias, ionospheric and tropospheric delay, and multipath due to reflections from nearby structures. RTK reduces these effects by operating on differences of observations, so that terms shared by the base and rover or by pairs of satellites cancel out. The core strategy is double differencing as shown in Fig. 1. Both receivers track the same set of satellites at the same time. A satellite-to-satellite difference at one receiver removes most of the receiver clock term. Forming a second difference between receivers removes most satellite clock errors and largely suppresses the slowly varying atmospheric components over short baselines. For instance, when two receivers track five satellites and one is designated as the reference, four independent double-difference equations are formed. This redundancy stabilises the solution and increases resilience to occasional outliers or brief signal degradations (Hofmann-Wellenhof, Lichtenegger & Wasle, 2008). These errors are mitigated using the differencing strategies outlined in the following section.

Because the carrier phase is measured in cycles, each observation contains an unknown integer number of whole wavelengths. Turning these very precise but ambiguous measurements into absolute coordinates requires ambiguity resolution (AR). During the 1990s AR was an active research topic, and algorithms such as LAMBDA became the standard way to fix the integers quickly and reliably. Today these methods are embedded in receiver firmware and run automatically once observations meet quality thresholds (Teunissen, 1995). AR is most reliable on short baselines and under a clear sky view, where residual atmospheric differences and multipath are small. As baselines increase, or when the environment is partially obstructed, ionospheric and tropospheric gradients, multipath, and orbit modelling errors significantly complicate the process of integer fixing. Modern systems address these challenges by choosing robust linear combinations of the carrier phases by improving initialisation procedures, and by continuously monitoring indicators such as residuals, fixed ratios, and the age of corrections. If a cycle slip is detected due to a temporary blockage, the rover can re-initialise and restore a fixed solution once signal quality returns (Rizos, 1997).

Modern RTK is typically dual-frequency (L1/L2), which accelerates ambiguity fixing and mitigates ionospheric delay. In cadastral use this yields:

The same technical principles translate into a practical operating checklist:

- Maintain PDOP ≤ 3 to ensure favourable satellite geometry.
- Track seven or more satellites distributed across at least two constellations so that the double-difference network remains well conditioned.
- Apply a sensible elevation mask so that very low satellites, which are more susceptible to multipath and atmospheric bending, are down-weighted or excluded.
- Monitor correction age within service limits and monitor quality indicators provided by the rover, including residual size; fix ratio and fixed/float status.
- For baselines beyond approximately 10–15 km, adopt ionosphere-aware combinations or use network RTK services so that spatially correlated errors remain small enough for fast and reliable integer fixing.

Following these controls enables the rover to continue delivering centimetre-level coordinates that are traceable, repeatable and suitable for boundary identification and mapping in cadastral applications (Hofmann-Wellenhof, Lichtenegger & Wasle, 2008; Leick, Rapoport & Tatarnikov, 2015; Rizos, 1997; Teunissen, 1995).

Why dual-frequency matters in practice. In cadastral surveys, dual-frequency RTK typically delivers:

- Faster fixes and re-fixes, so fieldwork moves continuously.
- Better resilience in urban sites or near vegetation where multipath and blockages occur.
- Lower ionospheric error, which helps on longer baselines.
- More consistent real-time accuracy, reducing downtime and rework across complex parcels.

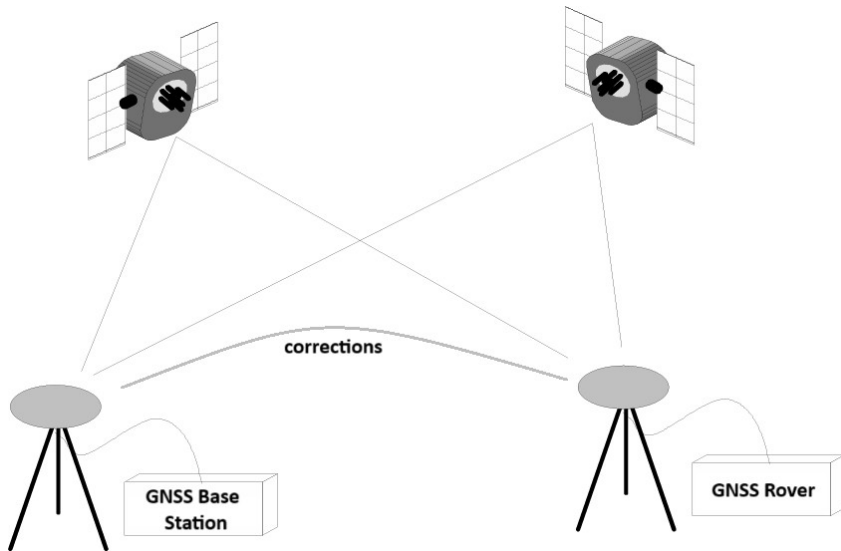


Figure 1: Double differencing

Ambiguity Resolution and Dual-Frequency in GNSS-RTK

GNSS receivers measure very fine changes in a radio wave called the carrier. Those measurements are incredibly precise, but they do not reveal how many *whole* waves lie between the satellite and the receiver. Ambiguity resolution (AR) is simply the process by which the equipment determines that whole-number count. Once resolved, the receiver can compute the position with centimetre-level accuracy. In the 1990s, researchers created fast, reliable ways to do this (for example, LAMBDA), and today those methods run automatically inside RTK receivers (Teunissen, 1995).

A simple picture of how AR helps

Imagine the receiver thinks the distance to a satellite is “ten thousand and a bit” wave lengths. The “and a bit” is easy; the hard part is the “ten thousand”. AR tests the possible whole-number answers and picks the one that best agrees with all the signals being tracked at the same time. When it picks the right one, the remaining error shrinks from a few centimetres to a few millimetres, and the rover reports a clean, fixed position.

When AR is easy, and when it is fussy

AR is usually quick when the rover is close to the base (short baseline) and the sky is reasonably open. In those conditions, the atmosphere and reflections from buildings or metal (called multipath) do not change much between base and rover, so the maths cancels them well. As the baseline grows beyond approximately 10–15 km, or when you work near walls, trees, or vehicles, the signals can be more disturbed, and fixing may slow down or drop back to float (not fully locked). Modern receivers deal with this by using smarter combinations of the signals, better “getting started” routines, and on-screen quality indicators that warn you when things are sliding, so you can pause or adjust and then re-fix quickly (Rizos, 1997).

Why two frequencies make life easier

Most current RTK gear tracks at least two frequencies (often called L1 and L2). That brings two big advantages:

- More information, quicker lock. With two frequencies the receiver has extra, independent data to test the whole-number answers, so time-to-first-fix and time-to-re-fix are usually shorter.
- Less trouble from the ionosphere. The upper atmosphere bends the radio waves slightly. Because it bends L1 and L2 differently, the receiver can compare them and remove much of that bending. This really helps on longer baselines or on ‘lively’ days when the ionosphere is active.
- A single-frequency rover can still reach centimetre accuracy if the baseline is short and the sky is kind, but it often needs post-processing back in the office. In the real world, where time matters—typical cadastral work—dual-frequency is preferred because it recovers faster after brief blockages (walking behind a van, tree or building corner), maintains fixed solutions for longer durations throughout the day, and preserves positional accuracy when the line of sight is not perfect (Leick, Rapoport & Tatarnikov, 2015).

Table 2 summarises the advantages of dual-frequency (L1/L2) tracking over single-frequency operation.

What this means in day-to-day surveying

Think of AR as the “click” that turns very good data into very reliable corner points. Dual frequency makes that “click” happen sooner, and more often, even under mixed conditions, such as transitioning from an open yard into a narrow alley. Over a whole traverse, staying fixed more of the time directly improves productivity and reduces repeat visits.

RTK survey procedure checklist:

- Keep the geometry healthy. Aim for PDOP ≤ 3 (your controller usually shows this). It means the satellites are spread out well enough for stable results.
- Watch the count. Try to track seven or more satellites from at least two constellations (e.g., GPS and Galileo). If one satellite drops out, you still have plenty to hold a fix.
- Mind low angles. Signals very close to the horizon are more likely to bounce and bend. Use an elevation mask of around 10–15°, or let the software down-weight low satellites automatically.
- Check the corrections. Keep the age of corrections small (your controller shows this in seconds). If it creeps up, pause or move for better mobile signal.
- Know your baseline. If you are more than 10–15 km from the base, use network RTK or the ionosphere-aware settings your gear provides. That keeps the “bending” effects small enough to fix quickly.

- Read the dials. Keep an eye on residuals, fixed/float status and any “ratio” or quality bar the rover shows. If residuals grow and the solution drops to float, stop, improve the antenna view, wait a few seconds, and let it re-fix.
- After a blockage. If you have just walked past a tree line or a vehicle and lost lock, stand still for a short moment. Most dual-frequency rovers will re-fix in place once the view clears.

Bottom line for cadastral jobs

Use LAMBDA-class ambiguity resolution for fast, stable fixes; keep an eye on the simple quality readouts and lean on dual-frequency to reduce ionospheric trouble. Shorten re-initialisation time and hold centimetre-level results throughout the day. With these habits, GNSS-RTK becomes a dependable tool for boundary identification and parcel corner work, not just in textbook conditions but in the varied, slightly messy settings most surveyors face (Teunissen, 1995; Rizos, 1997; Leick, Rapoport & Tatarnikov, 2015).

Feature	Single-Frequency RTK	Dual-Frequency RTK
Carrier Signal Used	Only L1	Both L1 and L2
Accuracy	2-5 cm (short baselines)	1-2 cm (short & long baselines)
Initialisation Time	Longer (30 sec to several minutes)	Faster (10-20 sec)
Ambiguity Resolution	Slower, requires stable conditions	Faster, more robust to errors
Multipath Resistance	Lower, more susceptible to signal reflections	Higher, reduces impact of multipath errors
Performance in Urban Areas	Poorer, more affected by signal obstructions	Better, can maintain lock even in challenging environments
Reliability for Cadastral Surveys	Adequate for open areas with short baselines	Ideal for complex environments and long baselines
Ionospheric Error Compensation	Not available errors must be modelled or corrected post-processing	Automatic correction using L1-L2 phase differences
Cost	Lower (€1,000-€3,000 per receiver)	Higher (€3,000-€10,000 per receiver)
Best Use Case	Short-range surveys in open environments	High-precision cadastral work in all environments

Table 2: *Comparison of Single-Frequency and Dual-Frequency GNSS-RTK*

GNSS-RTK Cadastral Measuring: A Practical Case Study at MCAST

Method

The Electronic Engineering Building at MCAST Kordin (Latitude: 35.877188, Longitude: 14.504178), as shown in Fig. 2, was selected as the primary site for cadastral GNSS-RTK measurement experiments. The rooftop was chosen for its unobstructed sky view, minimising non-line-of-sight (NLOS) effects and multipath, ensuring high-quality satellite reception suitable for accuracy testing in a controlled environment.

The experimental setup consisted of four main components: a GNSS base station, a GNSS-RTK rover unit, and a custom Android application for data collection and communication. In addition, a server was configured to relay real-time corrections from the base to the rover, completing the RTK chain.

Base station configuration: The base station was established at a known, fixed location on the rooftop and calibrated using traditional geodetic procedures over a continuous 24-hour session to ensure a stable, traceable coordinate. In-house hardware with multi-constellation GNSS chipsets continuously monitored satellite signals (GPS, Galileo, BeiDou and GLONASS). The base computed corrections for dominant error sources, including ionospheric delay—tropospheric delay, and satellite/receiver clock bias—and prepared them for transmission to the rover.



Figure 2: *Electrical and Electronics Block at MCAST Kordin*

Rover configuration and data logging: The GNSS-RTK rover was used to measure points, distances and areas on the rooftop test field. Built on the same in-house platform, the rover successfully tracked signals from over 20 satellites under typical observation conditions. Carrier-phase observations were recorded at 5 Hz to provide high temporal resolution for static and kinematic occupations. Data points (coordinates, distances, areas and status flags) were logged via the Android application so that measurement values and quality indicators were captured together for audit and post-processing.

Android application and visualization: A custom Android application interfaced with the rover, provided real-time feedback and supported on-device computations. The user interface employed Leaflet web mapping for visualisation, allowing operators to select points, verify fix status and view distances and areas calculated in real time (Joshi, Ghosh

& Singh, 2016). The layout prioritised clear buttons, map interaction and immediate access to measurement functions to reduce operator overhead during fieldwork.

Communications and server: A C# server running on a Windows desktop managed the data stream and relayed real-time corrections from the base to the rover. Communication with the rover was established via a 3G network using the HTTPS protocol to maintain low latency and secure transport. Measured average correction latency during trials was consistent with stable fixed-solution operation (as reported in Results), and the relay maintained continuity during routine movements and short-duration obstructions.

Error controls and quality assurance: Site selection and procedure design aimed to reduce multipath and improve solution stability. The rooftop location—together with standard error-handling routines in the base and rover firmware—limited reflections and supported robust ambiguity resolution. Visual inspection and software-based functionality tests were conducted before and during data collection to confirm system integrity. Any anomalies identified by app quality flags or post-collection checks were investigated and, where needed, corrected in post-processing.

Workflow Summary

- Establish the base on its calibrated rooftop mark and verify constellation tracking and correction output.
- Connect the rover; confirm fixed RTK status and nominal latency in the app.
- Acquire static occupations at control points and kinematic traverses along defined paths.
- Measure distances and areas for predefined shapes; verify on device; export logs with metadata.
- Review logs and quality indicators; repeat only where flagged; otherwise proceed to analysis.

This configuration—calibrated base, multi-constellation rover at 5 Hz, real-time corrections over 3G/HTTPS, and on-device visualisation with Leaflet (Joshi, Ghosh & Singh, 2016)—provided a consistent and auditable framework for centimetre-level cadastral measurements in the MCAST environment while maintaining traceability to ground control for the subsequent analysis and results.

Data Analysis and Processing

The raw data collected during the experiments passed through a processing pipeline designed to ensure accuracy, reliability and traceability. Real-time corrections flowed from the base to the server and then to the rover, enabling high-precision positioning in the field. To handle the raw data at the packet level, custom algorithms were used across low and high-level stages to ensure seamless communication and data integrity.

The primary technique for GNSS processing was double differencing, which eliminates major error sources such as satellite and receiver clock bias, ionospheric delay and tropospheric delay (Hofmann-Wellenhof, Lichtenegger & Wasl, 2008). In addition, a lightweight filter was applied to smooth residuals and stabilise the rover's position estimates without masking genuine changes. This combination allowed the system to achieve centimetre-level precision in practice.

Quality control combined visual inspection with automated checks. During the debugging stage, potential anomalies were flagged and addressed through targeted tests. The system was tested repeatedly to validate data integrity, and processed results were compared against ground-truth measurements to verify accuracy.

Raw observations—initially in hexadecimal packet format—were decoded using low-level coding techniques in strict adherence to established protocols. No assumptions were introduced during transformation. Each pipeline stage was unit-tested to maintain standards compliance and data integrity.

To assess robustness, measurements were taken at different locations on the rooftop, at different times of the day and across different seasons. The results showed the rover achieved 1 cm or better in static mode, confirming the expected precision and reliability for cadastral applications. The Android application provided real-time visualisation for immediate validation and efficient fieldwork (Joshi, Ghosh & Singh, 2016).

Positional Accuracy

Measured coordinates from the GNSS-RTK rover were compared against ground truth derived from traditional geodetic control, including manual triangulation and traverse methods established during the 24-hour calibration of the rooftop reference mark. Across multiple sessions and satellite geometries, the rover consistently achieved ≤ 1 cm horizontal error in static mode and 1–2 cm in kinematic mode. The Root Mean Square Error (RMSE) across repeated occupations averaged 0.8 cm in the horizontal plane and 1.2 cm in the vertical plane. These values were stable over different times of day and across changes in satellite geometry, indicating that the correction stream and ambiguity-fix persistence were sufficient for cadastral-grade positioning. Repeatability was assessed by occupying the same points on separate days; the standard deviation of coordinate differences across sessions was < 1 cm, confirming robustness for parcel boundary work.

Distance and Area Measurement

A series of distance and area measurements were performed using the rover and the Android application. Distances between predefined control points were compared with reference values obtained from traditional survey techniques; for short baselines a tape measure was used as an independent check where higher-precision instruments were unnecessary. The maximum distance deviation observed was 0.9 cm, with typical differences well below this threshold, indicating high reliability for boundary-line derivation and stakeout operations.

For area assessment, predefined geometric shapes were surveyed on the rooftop. The GNSS-RTK system accurately mapped areas, and the deviation between GNSS-RTK results and manually calculated reference values was $\leq 20 \text{ cm}^2$. While this deviation indicates high accuracy, it does not rule out that manual techniques may be less reliable than the GNSS-RTK approach. The observed difference could have originated from the manual method rather than from the GNSS-RTK system itself.

Real-Time Performance and Data Consistency

Real-time behaviour was evaluated by monitoring correction latency and solution stability. Corrections transmitted from the base to the rover via a 3G link (secured with HTTPS) were received with an average latency of 50–80 ms. This falls within acceptable bounds for sustaining a fixed RTK solution during typical field movement. During controlled interruptions (brief rover motion behind partial obstructions), re-initialisation was rapid, and solutions returned to fixed within practical limits for production workflows.

To test consistency, the same points were re-measured across multiple sessions, varying time of day and satellite geometry. The dispersion of solutions remained tight; standard deviation across sessions was < 1 cm, and no systematic drift was detected.

Error Analysis and Correction Effectiveness

Error sources considered included satellite geometry, ionospheric disturbance and multipath from nearby structures. Comparative trials with the rover operating without RTK corrections produced 2.5–4.2 m position errors, depending on geometry and site conditions. With real-time corrections and double differencing, positional deviations were reduced by approximately 90%.

Summary of Results

Table 3 summarises the measurement parameters and their respective accuracies, highlighting high precision in both static and kinematic modes. Key metrics such as RMSE, distance, and area errors confirm sub-centimetre accuracy with minimal correction latency.

Parameter	Measured Value
Static Mode Accuracy	≤ 1 cm
Kinematic Mode Accuracy	1–2 cm
RMSE (Horizontal)	0.8 cm
RMSE (Vertical)	1.2 cm
Distance Measurement Error	≤ 1.5 cm
Area Measurement Deviations	≤ 20 cm ²
Correction Latency	50–80 ms
Standard Deviation of Repeated Measurements	< 1 cm

Table 3: *Measurement Parameters and Accuracy*

Conclusion

The results of this study demonstrate the effectiveness of GNSS-RTK in achieving high-precision cadastral measurements. By integrating real-time corrections, robust error-mitigation techniques, and custom-built hardware and software, the system consistently delivered sub-centimetre accuracy. The findings confirm that GNSS-RTK provides a reliable and efficient alternative to traditional surveying methods, enabling faster and more precise land measurements. To extend performance to more challenging environments, future work will prioritise: (i) urban-canyon trials with quantified sky-view obstruction, multipath characterisation and ambiguity-fix persistence; (ii) operation under constrained connectivity using intermittent links and store-and-forward strategies; and (iii) resilience testing across variable baseline lengths and satellite geometries. Target metrics will include time-to-first-fix and re-fix, fixed-solution availability, horizontal/vertical RMSE, and latency under network loss and recovery. The author

intends to broaden the application of GNSS by developing advanced tools for robotics, Unmanned Surface Vehicles (USVs) and high-precision measuring applications. Planned work includes sensor fusion (e.g., GNSS with IMU, UWB or visual odometry), improved multipath mitigation, and AI-assisted quality control for real-time detection of non-line-of-sight and cycle slips. These capabilities aim to improve robustness and automation in positioning systems while maintaining centimetre-level performance. As GNSS-RTK technology evolves, its application space will continue to expand beyond land surveying to autonomous navigation, environmental monitoring and precision agriculture. By focusing on quantifiable robustness in difficult settings and on tightly integrated sensor platforms, the next phase of development can translate laboratory-grade accuracy into dependable performance across diverse operational scenarios.

Acknowledgments

This exercise was conducted as part of the preparation for the SmartPol project, which will utilise GNSS-RTK systems for Unmanned Surface Vehicle control in real time. Project SMARTPOL is funded by the MarTERA partners – Xjenza Malta and supported by the European Commission under grant agreement MRT-2021-057A.

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